What does a released average-subject brain encoder actually predict? A measured noise ceiling and a replication protocol

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Abstract

Released encoding models are increasingly used as instruments: a checkpoint is downloaded, run on new stimuli, and its predictions treated as a proxy for cortical response. That use depends on a property the release does not certify, namely that the specific checkpoint, in the specific configuration its loader selects, predicts held-out cortex at all. A self-published audit reports that a widely used trimodal encoder, in the average-subject configuration its loader forces, is *anti-correlated* with cortex at vertex level, $r = -0.0145$ against an inter-subject ceiling of $+0.0508$. The claim is about sign rather than magnitude, comes from a company selling the per-subject alternative, and has not been independently replicated. This paper does the part of the replication that requires no proprietary data and reports it separately from the part that does. From the public Algonauts 2025 responses we measure the leave-one-subject-out noise ceiling in parcel space for the episode a prediction pass would target: $r_{\text{ceiling}} = 0.1517$, 95% CI $[0.1484, 0.1551]$, over four subjects, all 1000 Schaefer parcels and 592 timepoints. This is the quantity any encoder must be judged against and it stands whatever the checkpoint turns out to do. We also report a withdrawn earlier value and the two defects that produced it, because both are silent failure modes that any group attempting this measurement can reproduce. The prediction pass itself is specified in full and marked as not yet run.

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1. Introduction

A released encoding model invites a particular kind of use: treat the checkpoint as an instrument, run it on stimuli its authors never considered, and read its predictions as a stand-in for measured cortical response. The appeal is obvious, since the alternative is scanning people. The risk is equally obvious once stated, which is that a technical report describing an architecture is not a warranty about the weights released under its name, and that a checkpoint may be released in a configuration whose predictive validity nobody has established.

For one widely used trimodal encoder that risk is not hypothetical. A commercial laboratory has published an audit of the released checkpoint in its average-subject configuration, reporting vertex-level $r = -0.0145$ against a measured inter-subject ceiling of $+0.0508$: anti-correlated with real cortex, across four subjects and all seven Yeo networks. Taken at face value, that would mean every downstream study using this checkpoint as an instrument is reading a signal with no established relationship to cortex.

The audit should not be accepted uncritically. It is self-published by a company selling the per-subject alternative, it uses four subjects and one stimulus, and it reports magnitudes near 0.015 in either direction, so it is a claim about sign rather than about effect size. Neither should it be dismissed. It is specific, it is falsifiable, and it concerns a configuration that a released loader selects by default.

This paper reports the half of the replication that public data supports, and specifies the other half rather than estimating it.

2. The measurement that stands on its own

An encoder's predictions are meaningless without knowing how much shared signal exists to predict. The leave-one-subject-out noise ceiling supplies that: each subject's response is correlated against the mean of the others, giving a lower bound on what any encoder could achieve, computed from the recordings alone with no model involved.

2.1. Data and procedure

We use the publicly released Algonauts 2025 responses, four subjects viewing the same naturalistic episode, parcellated into 1000 Schaefer parcels in MNI152NLin2009cAsym space. For each subject we correlate the parcel time series against the mean of the remaining three, per parcel, over time. Undefined correlations stay undefined: a parcel with no variance over time has no Pearson correlation, and substituting zero would pull the mean toward zero with invented values. Confidence intervals are percentile bootstrap over subjects with a fixed seed, so a reported interval reproduces exactly.

2.2. Result

For episode `s01e01a`, the episode a prediction pass would target,

$$r_{\text{ceiling}} = 0.1517, \qquad 95\%\ \text{CI}\ [0.1484,\ 0.1551], \tag{1}$$

over four subjects, 1000 of 1000 parcels defined, and 592 timepoints. Per-subject values are 0.1553, 0.1548, 0.1497 and 0.1470.

This is larger than the audit's +0.0508 by construction rather than by disagreement. Averaging vertices within a parcel cancels independent noise and raises correlations, so a parcel-level ceiling and a vertex-level ceiling are not comparable numbers, and conflating them would manufacture an agreement or a disagreement that is really a change of spatial unit.

3. Two silent failures, and why they are reported

An earlier version of this measurement gave 0.2247, CI $[0.2147, 0.2363]$, described as covering "482 parcels". That value is withdrawn. Both defects that produced it changed the number without raising an error, which is precisely why they are worth reporting: any group repeating this measurement can reproduce them.

The episode was selected by position. The analysis code carried a placeholder for the stimulus identifier and fell back to the first key in the response file. That key is a different episode from the one under prediction. Subjects do not agree on which session an episode is stored under, so positional selection is inconsistent across files as well. The ceiling was therefore measured on one episode while the prediction pass targeted another.

The array was transposed. The correlation routine expects (units, timepoints). The recordings are stored (timepoints, parcels), and the orientation guard tested whether the first axis was longer than the second. For a $(482, 1000)$ array that test cannot fire, so every correlation ran across the parcel axis: the reported quantity was an inter-subject spatial similarity per timepoint, presented as a per-parcel reliability. The "482 parcels" in the withdrawn figure was the timepoint count; the atlas has 1000 parcels.

Running the original code path against the same four files reproduces 0.2247, CI $[0.2147, 0.2363]$ and the count of 482 exactly, which is what identifies the cause rather than merely fitting it.

Two practices follow, and we state them as recommendations because both defects are generic. Select records by name, never by position. Decide array orientation against a known constant such as the atlas size, never by comparing axis lengths, since an acquisition with more timepoints than units inverts a length-based rule and still produces a plausible correlation.

4. The prediction pass, specified and not yet run

The remaining half of the replication requires running the checkpoint over the same stimulus and correlating its predictions against the same recordings. That pass is written and is not yet run: the compute budget available for this work was exhausted by the corpus scan described in the companion paper. It is specified here in full so the claim of what remains is checkable rather than vague.

1. Load the released checkpoint in the average-subject configuration its loader forces, which is the configuration the audit concerns.
2. Predict the response to episode `s01e01a`, the episode whose ceiling is reported above.
3. Project predictions from the encoder's fsaverage5 vertices into the 1000 Schaefer parcels, rather than upsampling the recordings, since upsampling would invent within-parcel structure the data does not have.
4. Match the recording to the prediction by episode name, and assert orientation against the parcel count, for the reasons in Section 3.
5. Correlate per parcel over time, and report the sign as measured. Nothing is flipped or absolute-valued: the sign is the finding under test.

The comparison to state on completion is between the encoder correlation and the ceiling above. A negative encoder correlation against a positive ceiling would replicate the audit's direction in parcel space. A positive one below the ceiling would mean the checkpoint carries signal but less than people share with each other. Neither result is comparable with the audit's vertex-level -0.0145 without vertex-level surface derivatives that the public mirror does not carry.

5. Conclusion

The noise ceiling for the target episode is 0.1517, CI $[0.1484, 0.1551]$, measured from public data and independent of any model. It is reported here because it stands on its own and because the prediction pass it exists to calibrate is not yet run. The withdrawn value and its two causes are reported for the same reason a null is: both are silent, both are reproducible by anyone attempting this measurement, and neither would have announced itself.

Code and data availability

The measurement script, the withdrawn code path, and the artifact recording every value above are public at `https://github.com/brn-mwai/monarch`. The ceiling is produced by `scripts/paper3_noise_ceiling.py`, which takes the episode as an argument, matches the record by name and asserts orientation against the atlas size.

CRediT authorship contribution statement

Brian Mwai: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization. The author is the sole contributor to this work.

Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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